School of Engineering & Technology

Department of Computer Science & Technology

AUTONOMOUS RESEARCH AGENT USING LANGCHAIN

Agentic AI Lab (CSCR3215)
ASSIGNMENT-2
B. Tech. (CSE)
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Autonomous Research Agent using LangChain for Automated Topic Research and Report Generation

Introduction:

An Autonomous Research Agent is an intelligent software system designed to automatically collect, analyze, and summarize information related to a given topic. Instead of manually searching multiple websites and reading large amounts of content, the agent performs these tasks automatically using artificial intelligence.

In this project, LangChain is used to connect a Large Language Model with external tools such as web search and Wikipedia. The user provides a research topic, and the system gathers relevant information from different sources, processes the collected content, and generates a structured report.

This project demonstrates the practical use of AI agents in automating research-related tasks and reducing human effort in information collection.

Objective:

The main objective of this project is to design an AI-based research assistant capable of automatically exploring a topic and generating a detailed report.

Specific objectives include:

- Accepting a research topic from the user
- Searching relevant information automatically
- Extracting useful knowledge from trusted sources
- Summarizing collected information
- Organizing final output in report format

Problem Statement:

Traditional research methods require users to manually search websites, read multiple documents, compare information, and prepare summaries. This process consumes significant time and effort.

To solve this issue, an Autonomous Research Agent is developed using LangChain. The agent combines reasoning capability of language models with external tools to automatically perform information gathering and report generation.

Key Findings:

• LangChain enables integration of language models with external tools.

• Web search tool helps retrieve current and relevant information from online sources.

• Wikipedia tool provides structured background knowledge for the selected topic.

• The agent combines data from multiple sources before generating final output.

• Automated report generation improves speed and reduces manual research effort.

Challenges:

- API key setup required careful configuration.
- LangChain version differences created import errors during implementation.
- Combining outputs from multiple tools required proper formatting.
- Some search results needed filtering before summarization.
- Dependency installation caused version compatibility issues.

Future Scope:

- Integration of PDF document research tools.
- Addition of multiple search engines for richer data collection.
- Development of web interface for easier user interaction.
- Automatic citation generation in reports.
- Real-time research updates using live internet sources.

Conclusion:

The Autonomous Research Agent successfully demonstrates how AI can automate research tasks using LangChain and external tools. The system accepts a topic, collects information, analyzes content, and produces a structured report.

This project shows practical application of intelligent agents in modern AI systems and highlights the potential of automated research assistants in academic and professional environments.

Sample Output 1

Topic:
Impact of AI in Healthcare

Introduction:
Artificial Intelligence improves diagnosis, treatment planning, and hospital management.

Key Findings:
• Better disease detection
• Faster decision support
• Improved robotic surgery

Challenges:
• Privacy concerns
• High implementation cost

Future Scope:
• Personalized medicine

Conclusion:
AI will significantly improve healthcare systems.

Sample Output 2

Topic:
Blockchain in Education

Introduction:
Blockchain secures educational records and digital certificates.

Key Findings:
- Secure certificate verification
- Tamper-proof records

Challenges:
- Adoption complexity

Future Scope:
- Global verification systems

Conclusion:
Blockchain can improve academic transparency.